The depth exchange rate: what index attention computes, layer by layer (expansion note, 2026-10-20)

Scope: main draft Section 5 (`docs/paper/MAIN_DRAFT1.md`), outline claims 5.1-5.4 (`docs/paper/OUTLINE_main_v2.md`). Every statement carries one label:

- **ESTABLISHED** -- a registered result in a results file (the results file is authoritative; quoted, not re-derived).
- **ANALYSIS** -- computed here, post hoc, eval-only on stored checkpoints (CPU), after the batches were read. Not pre-registered; descriptive. Each number names its script and output file.
- **THEORY** -- an argument (a construction, an upper bound, or a cited theorem applied to our setting). Not a measurement.
- **SPECULATION** -- a mechanism the data are consistent with but do not establish.

Scripts and outputs: `docs/paper/expansion/scripts/depth/` (outputs in `out/`). Every stored accuracy was reproduced before any new readout was taken (Section 4.0). No GPU was used; nothing was trained.

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1. Summary

1. **ESTABLISHED (the rates).** One path layer is worth about three index layers on Dyck-2 at matched depth and on the biased ring wherever a step can be undone; one index layer suffices when the walk never reverses (p_plus 1.0); on the torus at T = 1024 four index layers or a loop of four fall far short (0.860 / 0.864 vs 1.000), and depth helps only short-gap revisits.
2. **THEORY (why path integration needs one layer).** A revisit at move t must retrieve an observation at any earlier s with P_s = P_t (mod N), P the prefix sum of the moves. In the path model the score depends on omega (P_t - P_s) directly: the integrator supplies the prefix sum, exactly and for any gap, before attention. Index attention sees position only as t - s, so the predicate "the moves in (s, t] sum to zero" must be computed in the residual stream.
3. **THEORY (two routes for index attention).** (A) *Global count*: uniform causal attention averages the moves, giving P_t / n_t; rescaled and matched, this is a depth-2 solution in principle (Liu et al. 2023, Lemma 6; Yao et al. 2021's depth-2 Dyck construction), but it needs a normaliser and precision 1/n_t, its MLP width and weights grow with N x T in the known construction, and the target is highly sensitive (every one of the g moves since the last visit decides it; Hahn and Rofin 2024). Its cost grows with T, not with depth. (B) *Local return set*: a layer can realise rules whose return offset depends on a bounded number of recent direction runs (*segments*): undoing the last run (retrace), or a constant offset (a clock). Its cost grows with the number of segments a revisit spans, and depth extends that number.
4. **ANALYSIS (route B is what deep index attention learned on the torus).** Of the torus's short-gap revisits, 82% are one-level retraces; four index layers or the loop solve them (0.996 / 1.000), and on the rest accuracy falls with the number of segments spanned: 0.93 / 0.98 at 3-4 segments, 0.74 / 0.65 at 5-8, 0.56 / 0.50 at

9-16, floor beyond. At a fixed gap of 32-127 moves the same fall appears (0.87 / 1.00 at 2 segments to 0.49 / 0.50 at 17+), so segments, more than the gap in moves, limit the deep models (Section 4.4 finds a smaller residual gap effect). This accounts exactly for "just above retrace": 0.943 = (0.680 x 0.996 + 0.148 x 0.699) / 0.828. Their last layer puts 0.55-0.58 of its attention on the last visit's observation for retrace queries, 0.22 for other short gaps, and 0.00-0.01 for long-gap and wrap-only revisits (path model: 0.81 / 0.74 on long gaps). Linear probes find no global count in their residual stream (unwrapped displacement R^2 <= 0.10; normalised count P_t / t <= 0.21), and a displacement code over the recent window that degrades with the window (R^2 0.96 at 4 moves, 0.77 at 64, 0.37 at 256).

5. **ANALYSIS (route A is what three index layers learned on the ring).** On the ring (T = 128, fresh seeds 2-7), one index layer learns exactly two rules: retrace (0.68-0.80) and, when the walk is biased, the one-lap clock (0.987-1.000); every other revisit is at the floor (0.49-0.59). Two layers add segment composition that decays with segments (0.89 / 0.80 / 0.68 at 3-4 / 5-8 / 9-16 segments, p_plus 0.5). Three layers are flat in segments (0.997 / 0.993 / 0.972) and their residual stream carries a global count (P_t / t R^2 0.69, P_t 0.60, against 0.18 / 0.09 at one layer): the count route is found on the ring and not on the torus.
6. **ANALYSIS (the 0.75 dip, descriptive only).** One-layer index accuracy across p_plus is the share-weighted sum of those two rules and the floor: a1 = 0.709 / 0.678 at p_plus 0.5 / 0.75 is reproduced from class shares and class accuracies. The dip is the retrace share falling (0.72 -> 0.49) before the clock's share is large (0.18). The pre-registration attached no mechanism sentence to the dip; this is a post hoc decomposition, not a verdict.
7. **ANALYSIS (Dyck).** One index layer predicts a closer only when its opener is the previous bracket (0.998), and is near chance (0.61-0.64) once one balanced block intervenes; each further layer extends the number of blocks it skips (two blocks: 0.78 / 0.89 / 0.93 at 2 / 3 / 4 layers), with a residual dependence on distance. Dyck has a BOS token, so the count route's normaliser is available; the data are a mixture and do not separate the routes cleanly.
8. **Proposed paper text** (Section 5) and **five GPU experiments** that test the account (Section 6). The decisive one varies the walk's run length at fixed T: route B predicts accuracy is a function of segments spanned, route A a function of T and the gap. A CPU eval-only version on the stored models (Section 4.4) agrees: retraces are solved at any gap, and at a fixed gap of 16-63 moves 4L scores 0.98 / 0.87-0.93 / 0.72-0.80 / 0.64-0.70 at 2 / 3-4 / 5-8 / 9-16 segments, with a smaller residual dependence on the gap.

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2. What is being explained (ESTABLISHED)

task (trained and tested at the same length / depth)	path, 1 layer	index 1 / 2 / 3 / 4 layers	rate	file
Dyck-2, L32 D12, A2f (floor 0.594)	0.996	0.648 / 0.889 / 0.964 / 0.982 (RoPE)	~3	`DYCK_MDEPTH_RESULTS.md`
ring 32, T = 128, p_plus 0.5 (fresh seeds 2-7)	1.000	0.710 / 0.949 / 0.998	3	`CANCEL_RESULTS.md`, `docs/audits/2026-09-27/fresh_seeds.txt`

task (trained and tested at the same length / depth)	path, 1 layer	index 1 / 2 / 3 / 4 layers	rate	file
ring, p_plus 0.75 / 0.9	1.000	0.677 / 0.962 / 0.998; 0.824 / 0.987 / 0.999	3 (0.9 knife-edge)	same
ring, p_plus 1.0	1.000	1.000	1	same
torus 64x64, T = 1024 (n = 4)	1.000	0.719 / -- / -- / 0.860; loop x4 0.864	> 4	`FACTORIAL_RESULTS.md`

Torus strata (registered secondary; floors post hoc, `docs/paper/figs/fig_factorial_strata_out.txt`): index 1 layer / 4 layers / loop: short gap 0.769 / 0.943 / 0.938 (retrace floor 0.912), long gap 0.480 / 0.458 / 0.507, wrap-only 0.482 / 0.457 / 0.509 (constant floors 0.515 / 0.517). Every index run STALLED; budget-scoped (900 epochs torus, 300 ring, 1x Dyck ladder budget; at 3x budget four index layers reach Dyck's ceiling, 0.997-0.998).

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3. Theory

3.1 What a revisit prediction requires (THEORY, elementary)

Tokens interleave moves and observations, [a_1, o_1, a_2, o_2, ...]; o_t is predicted at the token a_t. Let P_t be the unwrapped displacement after move t (an integer in 1D, a pair in 2D). The map is redrawn per evaluation, so the only source of o_t is an earlier observation token o_s at the same cell: P_s = P_t (mod N per axis). Because the map is fixed within a walk, *any* earlier visit will do; no recency tie-break is needed.

A causal attention score between query t and key s in an index-RoPE layer is

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score(t, s) = q(h_t)^T R((t - s) Omega) k(h_s),
```

so position enters only through the offset t - s, and the content h_t, h_s are the residual streams. The target predicate P_t - P_s = 0 (mod N) is a property of the *interval* (s, t]. There are two ways to make the score depend on it:

- **(A) absolute code**: write a function of P_t into every h_t, so that q(h_t)^T k(h_s) peaks at P_s = P_t (mod N). This requires the prefix sum P_t in the residual stream.
- **(B) relative return set**: write into h_t a description of the set of offsets d <= W at which the recent path returns to the current cell, and let the RoPE term select those offsets: score(t, t - d) = sum_c Re(alpha_c(t) e^{i omega_c d}) approximates the indicator of the return set on [1, W]. This needs only the last W moves, but the query must encode their return structure.

3.2 Path integration supplies the prefix sum (THEORY, exact)

MapWM rotates queries and keys by theta_t = omega * cumsum(Delta(x)). With observation steps near zero and move steps a fixed vector per move (MapFormer's learned solution), theta_t = omega P_t and score(t, s) = q^T R(omega (P_t - P_s)) k: route

(A) with the prefix sum computed outside attention, exactly (a float32 cumsum), for every gap and every T. In a channel whose frequency times the learned move step is a multiple of 2 pi / N, the score is N-periodic in P, and wrap-around costs nothing (the slowest default frequency has period N moves at unit step; Prop. 1 of the draft concerns which step tables represent this). One layer suffices; the stored path models show it (same-cell attention 0.81 / 0.74 on long-gap torus revisits, 0.66 / 0.58 on wrap-only; Section 4.2).

3.3 Route A for index attention: depth 2 in principle, precision in practice (THEORY)

Construction (upper bound on depth). Layer 1: one head with constant query and key attends uniformly to the prefix; its value carries the move vector on move tokens, so its output is P_t / n_t with n_t = 2t the prefix length. A second feature proportional to 1 / n_t is available in the same layer (for example, a head with a fixed self-attention bonus beta has self-weight e^beta / (e^beta + n_t - 1)). The MLP rescales to P_t and forms periodic features cos(2 pi k P_t / N), sin(2 pi k P_t / N). Layer 2 matches q(h_t)^T k(h_s) on those features. This is Liu et al.'s (2023) Lemma 6 (one layer simulates a mod-n counter: prefix average + MLP mod) followed by a match, and Yao et al.'s (2021) depth-2 Dyck construction (counter in layer 1, proximity-based depth matching in layer 2; as summarised in Liu et al., App. A.5). In RASP terms the task has depth 2 in 1D and in 2D alike (two counts are two value dimensions of one head) and with or without wrap (the mod is in the MLP). **So the measured rates (3, 3, > 4) are not representational depth bounds.**

What the construction costs. - A normaliser. Lemma 6 uses a padding token and an absolute position channel gamma_t = log(2T - t) to undo the 1/t of causal averaging. Our torus and ring sequences have neither (no BOS; RoPE is relative); Dyck has a BOS. The model must improvise the 1/n_t feature (it does: 1/t is linearly decodable at R^2 0.67-0.89 at the best pass of every multi-layer index model, Sections 4.3, 4.5). - *Precision.* Adjacent cells differ by 1/n_t in the averaged code: 1/256 at the end of a ring walk (T = 128), 1/2048 per axis at the end of a torus walk (T = 1024). - *Width and weights.* Lemma 6's mod gate uses MLP width 4 n T and weights up to 4 n T + 2 (width O(n) only with periodic activations). Read as an upper-bound construction: 32 x 256 for the ring, 64 x 2048 per axis for the torus (16x per axis, two axes); our MLP has width 512. - *Sensitivity.* For a revisit at gap g, flipping any one of the g moves in (s, t] moves the target cell, while flips before s do not; the prediction's sensitivity grows with g. Hahn and Rofin (2024, Cor. 5 and Thm. 6) show that representing a function whose average sensitivity grows faster than sqrt(n) log n needs parameter norms or layer-norm blow-up that grow with input length, and that such minima are sharp. Long-gap torus revisits (g in the hundreds against sqrt(n) ln n ~ 345 at n = 2048 tokens) approach that regime; short-gap revisits (g < 16 moves) are far from it. This cost does not by itself separate the ring from the torus (ring gaps reach 128 moves against sqrt(n) ln n ~ 89 at n = 256); precision, width and the second axis do. Their theorem is for Boolean inputs under the uniform distribution; we use it for direction only.

Prediction. Route A is found when T and the dimension are small and a normaliser is easy (ring at T = 128; Dyck with a BOS at length 32) and not when T is large (torus at T = 1024). Where it is found, accuracy does not depend on how many segments a revisit spans.

3.4 Route B: bounded return rules, extended by depth (THEORY, with an empirical reach)

One layer. Two rules need no prefix sum. - *Retrace.* While the current run of moves reverses the previous run, the walk returns along its own track: after j steps of reversal the cell is the one seen 2j moves (4j tokens) earlier. The query must encode j (the current run's length, a local average over the last <= 10 moves) and rotate its RoPE phases by the matching offset (4j tokens). This is a bounded-window rule (runs are at most 10 moves); its sensitivity is bounded by the two runs involved. - *Clock.* When the walk never reverses (p_plus 1.0) P_t = t, and the cell visited N moves earlier is the current cell: a constant offset, which RoPE represents in one head. Position equals index; there is nothing to integrate.

Depth. A return that spans k segments (maximal runs of one action) is a function of those k segments' directions and lengths. To select its offset the query needs a summary of k segments; segment boundaries sit at content-dependent offsets, so fetching the summary of the segment before the last is a pointer hop. With two heads a layer can make a bounded number of hops; stacking layers extends the reach. Two bounds bracket the reach: hops compose linearly if each layer adds one segment, and logarithmically if layers double their pointers (Liu et al.'s Theorem 1 gives O(log T) depth for any semiautomaton by such doubling; for an Abelian group like ours Lemma 6's depth-1 route exists, which is route A). We have only L = 1 and 4 on the torus, so the shape (linear or logarithmic) is open (experiment E2).

Prediction. At fixed depth, accuracy falls with the number of segments spanned, and depends on T or the gap only through them. The depth needed to solve a task is set by the segment counts of its revisits: on a 2D walk at T = 1024, long-gap revisits span 17 or more segments in 98.5% of cases (median 37; wrap-only: 99%, median 65; Section 4.1), far beyond what 4 layers reach.

3.5 What the two routes predict for each rate (THEORY, checked in Section 4)

task	route B covers	route A	predicted rate	checked
ring, p_plus 1.0	everything (clock: constant offset)	not needed	1	4.5: lap class 1.000 at 1 layer
ring, p_plus 0.5-0.9, T = 128	retrace + clock at 1 layer; 3-8 segments partly at 2	ring is small: count learnable	3 (count found at 2-3)	4.5: 3 layers flat in segments, count decodable
Dyck-2, L32 D12	adjacent opener at 1 layer; blocks to skip grow with depth	BOS available	~3-4	4.6: block dependence shrinks with depth
torus, T = 1024	retrace; ~4 segments at 4 layers	not found	> 4, growing with segments	4.1-4.4

3.6 Assumptions and limits of the theory

- Route A's depth-2 construction is an upper bound on the depth needed and an existence statement; it says nothing about what SGD finds. Route B's reach is

measured, not derived; the per-layer hop model is a SPECULATION about the mechanism, consistent with the attention and probe data.
- The two routes are not exhaustive (a model can mix them, as on the ring at 2 layers and on Dyck).
- The sensitivity argument transfers Hahn and Rofin's Boolean, uniform-input theorem to structured walks; it predicts a direction (long gaps harder), not a threshold.
- Linear probes bound what attention can read linearly from LN(h); a nonlinear code could exist undetected. The torus models' retrieval failure at long gaps (attention mass 0.00-0.01 on the cell) does not depend on the probes.

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4. Checks run (ANALYSIS; post hoc, eval-only, CPU)

4.0 Reproduction (every check passed before any new readout)

- Torus (`torus_depth.py`): the FACTORIAL eval stream re-generated (env seed 10000, np seed 1234 + s, 100 walks); overall accuracy per run equals `FACTORIAL_{L1,L4,LOOP}_STRATA.json` "all" to 4 decimals for all 16 runs; the re-implemented forward (to expose attention and residuals) equals the model's logits (max |diff| 0.0).
- Ring (`ring_depth.py`): the CANCEL eval stream (train_cancel.evaluate: np seed 0, env seed 10000, 200 walks); overall accuracy equals each run's `eval.json` "128" for all 128 runs (asserted < 2e-3; printed values equal to 4 decimals). Only fresh seeds 2-7 are read below.
- Dyck (`dyck_distance.py`): A2f recomputed per run; arm means equal `DYCK_MDEPTH_RESULTS.md` (RoPE 0.648 / 0.889 / 0.964 / 0.982, PoPE 0.640 / 0.850 / 0.946 / 0.971, MapWM 0.996 / 1.000).

4.1 Torus: which revisits depth buys (`torus_depth.py`, `summarize_torus.py` -> `out/torus_depth_out.txt`)

Per revisit: s = the last visit to the cell; *retrace* = the current run reverses the previous one and has not overshot it (then s = t - 2 cur, the one-level retrace rule of the floors); *segments* = maximal single-action runs among moves s+1..t. Mean of per-seed accuracies (n = 4 seeds, range in brackets); share = fraction of all revisits.

revisit class	share	index 1L	index 4L	loop x4	path 1L
short gap, retrace	0.680	0.833 (0.79-0.86)	0.996 (0.99-1.00)	1.000	1.000
short gap, not retrace	0.148	0.475	0.699 (0.51-0.77)	0.655	1.000
... 3-4 segments	0.031	0.477	0.926 (0.74-0.99)	0.977	1.000
... 5-8 segments	0.056	0.476	0.735 (0.52-0.81)	0.645	1.000
... 9-16 segments	0.052	0.474	0.561 (0.38-0.64)	0.499	1.000
... 17+ segments	0.009	0.472	0.493	0.502	1.000
long gap (>= 128)	0.090	0.480	0.458	0.507	1.000
wrap-only	0.083	0.482	0.457	0.509	1.000

Gap in moves (plain revisits): index 4L / loop 1.000 / 1.000 at 2-7, 0.978 / 0.986 at 8-31, 0.614 / 0.551 at 32-127, 0.458 / 0.506 at 128-511. One layer: 0.935 at 2-7, 0.681 at 8-31, floor beyond.

Segments at a fixed gap (plain revisits, gap 32-127 moves; pooled over seeds):

segments spanned	n	index 1L	index 4L	loop x4
2	240	0.412	0.874	0.996
3-4	503	0.483	0.832	0.911
5-8	4565	0.464	0.688	0.575
9-16	6569	0.474	0.560	0.499
17+	1202	0.472	0.493	0.502

At gap 8-31 the same: 4L 0.993 / 0.940 / 0.818, loop 1.000 / 0.987 / 0.768 at 2 / 3-4 / 5-8 segments.

Accounting for "just above retrace". The short-gap retrace floor is (0.680 x 1 + 0.148 x ~0.5) / 0.828 = 0.911 (measured 0.912). Four index layers: (0.680 x 0.996 + 0.148 x 0.699) / 0.828 = 0.943 (measured 0.943). Depth adds the exact retrace rule plus partial returns over 3-8 segments; that is the whole of its short-gap margin over retrace.

Long-gap and wrap-only revisits span many segments: 98.5% of long-gap revisits span 17 or more (median 37, minimum 11), and 99% of wrap-only revisits (median 65). They are *below the constant floor* (0.458 / 0.457 vs 0.515 / 0.517) with high NLL (2.1-4.8 in `FACTORIAL_L4_STRATA.json`): the deep models predict a specific wrong object rather than abstaining to blank.

4.2 Torus: where index attention looks (same script)

Attention at the revisit query (token a_t), per pass and head (h0 / h1), mean over revisits and seeds. *same* = mass on earlier observation tokens of the same cell (uniform-attention chance 0.001-0.003); *last* = mass on the last visit's observation and arrival tokens.

model, final pass	retrace: same / last	short, not retrace: same	long gap: same	wrap-only: same
index 1L	0.05 / 0.10 (h0), 0.06 / 0.12 (h1)	0.00	0.00	0.00
index 4L, layer 4	0.55 / 0.56, 0.58 / 0.58	0.22 / 0.22	0.01 / 0.01	0.00 / 0.00
loop x4, pass 4	0.50 / 0.52, 0.49 / 0.50	0.20 / 0.20	0.00 / 0.00	0.00 / 0.00
path 1L	0.83 / 0.70, 0.76 / 0.65	0.82 / 0.76	0.81 / 0.74	0.66 / 0.58

In the deep index models retrieval happens in the last pass only (layers 1-2: same-cell mass <= 0.04 everywhere; layer 3 0.21-0.33 for retrace), and on retraces all of it is on the last visit (same = last): they find the mirror point, not the cell. Split by outcome, the final pass's max-over-heads same-cell mass is 0.75 on correct and 0.30 on wrong retrace predictions (4L), 0.45 vs 0.21 on other short gaps, and 0.02 vs 0.02 on long gaps: at long gaps there is no retrieval to get right or wrong.

4.3 Torus: no global count (`torus_probe.py` -> `out/torus_probe_out.txt`; `torus_depth.py` probes)

Ridge probes from the residual stream at move tokens (train walks 0-29, test 30-39, t >= 16; R^2 out of sample, mean of 4 seeds, x and y averaged). Best pass shown.

target	index 1L	index 4L	loop x4	(ring index 3L, p_plus 0.5, for comparison; 4.5)
unwrapped displacement P_t	-0.06	0.10	-0.01	0.60
normalised count P_t / (t+1)	-0.03	0.21	0.06	0.69
1 / (t+1)	-0.01	0.78	0.74	0.89
displacement over the last 4 moves	0.90	0.96	0.95	0.97
... last 16	0.47	0.93	0.86	0.86
... last 64	0.09	0.77	0.56	0.70
... last 256	-0.01	0.37	0.14	--
torus phase cos / sin (2 pi P / 64)	-0.02	-0.01	-0.01	(ring phase 0.04-0.10)

The deep torus models know elapsed time (1/t) but carry no normalised count; they carry a recent-displacement code whose error in cells grows with the window (rmse = sd x sqrt(1 - R^2) with the window sds of `out/walk_stats_out.txt`: 0.5 cells at 4 moves, 1.8 at 16, 7.1 at 64, 25 at 256 for 4L). That is route B's signature. (The path model's residual stream carries no position either, as expected: its position is in the rotation.)

4.4 Torus: run length moved at evaluation (`torus_runlen.py` -> `out/torus_runlen_{L1,L4,LOOP,PATH}_out.txt`)

The stored FACTORIAL models re-evaluated (no training) on walks whose run length is fixed at 2, 5 or 10 moves instead of drawn from 1-10 (within the training support of run lengths, but a shift in walk statistics; held-out map; 60 walks per seed for 1 layer, 30 for 4L / loop, 10 for path). Plain revisits, pooled over the 4 seeds; cells with >= 50 revisits. Rows: segments spanned; columns: gap in moves (2-15 / 16-63 / 64-127 / 128+). Index 4 layers (loop in brackets):

condition	2 segments (retrace)	3-4	5-8	9-16	17+
training walk (1-10)	0.999 / 0.978 (1.000 / 1.000)	0.978 / 0.925 (0.969 / 0.978)	0.875 / 0.759 / 0.527 (0.838 / 0.675 / 0.418)	-- / 0.635 / 0.539 (0.512 / 0.505)	0.440 / 0.459 (0.481 / 0.509)
runs of 2	0.999 (1.000)	0.934 / 0.898 (0.884 / 0.966)	0.836 / 0.783 (0.744 / 0.621)	0.702 (0.526)	0.580 / 0.524 / 0.461 (0.483 / 0.489 / 0.493)
runs of 5	0.997 / 0.979 (1.000 / 1.000)	0.997 / 0.912 (0.999 / 0.982)	0.803 / 0.819 (0.683 / 0.511)	0.701 / 0.654 (0.517 / 0.503)	0.551 / 0.465 (0.509 / 0.509)
runs of 10	1.000 / 0.978 (1.000 / 0.995)	0.867 / 0.662 (0.939 / 0.382)	0.715 / 0.605 (0.556 / 0.503)	0.557 / 0.449 (0.511 / 0.494)	0.453 (0.506)

(Each cell lists the populated gap columns left to right.) Overall plain accuracy of 4L: 0.901 on the training walk, 0.843 / 0.858 / 0.841 with runs of 2 / 5 / 10; path 1.000 in every cell of every condition.

- *Two-segment returns are gap-invariant for deep index models*: retraces at gaps of 16-63 moves (runs of 10) score 0.978 / 0.995. One index layer's retrace is window-limited instead: 0.83-0.93 at gaps of 2-15 moves and 0.44-0.46 at 16-63 in every condition (`torus_runlen_L1_out.txt`).
- *Segments set the level, the gap trims it*: down a column of fixed gap (16-63 moves) 4L falls from 0.98 (2 segments) to 0.87-0.93 (3-4), 0.72-0.80 (5-8), 0.64-0.70 (9-16) and 0.58 (17+, runs of 2); within a segment bin across gaps it moves by 0.02-0.2 (0.35 in one cell of 55 revisits). The loop is sharper: 0.88-1.00 up to 4 segments at gaps below 64, at or near the floor (0.50-0.53) from 9 segments at any gap.
- Shorter runs put more segments into each gap and lower overall accuracy (0.843 with runs of 2), as route B predicts. Runs of 10 also lower it (0.841), through 3-8 segment returns at gaps of 16-127 moves (4L 0.60-0.87), so the gap is not irrelevant.
- Reading: route B with a reach of about 4 segments at 4 layers (partial to 8), plus a weaker gap dependence. Because the conditions shift the walk distribution, this is suggestive; E1 (Section 6) trains on each condition.

4.5 Ring: rules, segments and a count (`ring_depth.py`, `summarize_ring.py` -> `out/ring_depth_out.txt`)

Classes: *retrace* as on the torus; *lap* = the displacement over the last 32 moves is 0 mod 32 (a fixed one-lap offset lands on the cell; the clock); *neither* = the rest, split by segments. Fresh seeds 2-7, mean (range).

p_plus 0.5:

class	share	index 1L	index 2L	index 3L	path 1L
retrace	0.716	0.797	0.987	1.000	1.000
lap, not retrace	0.035	0.497	0.977	1.000	1.000
neither, 3-4 segments	0.127	0.487	0.891 (0.72-0.94)	0.997	1.000
neither, 5-8	0.097	0.489	0.802 (0.55-0.87)	0.993	1.000
neither, 9-16	0.025	0.485	0.678 (0.48-0.75)	0.972	1.000
wrap-only	0.118	0.503	0.819	0.989	1.000

p_plus 0.75 / 0.9 (index 1L): retrace 0.692 / 0.677, lap 0.987 / 0.994, neither 0.487 / 0.592. p_plus 1.0: every revisit is a lap revisit, 1.000 at every depth.

- One index layer learns exactly retrace and (when it is common) the clock, and nothing else.
- Two layers compose segments, decaying with their number (route B).
- Three layers are nearly flat in segments, and the residual stream at layer 2 carries a global count: P_t R^2 0.60, P_t / t 0.69, 1/t 0.89 (one layer: 0.09 / 0.18 / 0.14; two: 0.40 / 0.60 / 0.67). At p_plus 0.5 there is no drift, so P_t is not a proxy for t here. Route A is found on the ring.

- Its last-layer attention reaches the cell on wrap-only revisits (same-cell mass 0.36 / 0.38) and other non-local revisits (0.17 / 0.18), where 1-2 layers do not (0.00-0.14). The probes find no linear phase code (cos / sin R^2 <= 0.10), so how the count is matched across the wrap is not identified.

The non-monotone a1(p) (descriptive). With per-class accuracies at each p, a1 = sum over classes of share x accuracy: p_plus 0.5: 0.716 x 0.797 + 0.035 x 0.497 + 0.249 x 0.487 = 0.709 (observed 0.710); 0.75: 0.493 x 0.692 + 0.179 x 0.987 + 0.329 x 0.487 = 0.678 (0.677); 0.9: 0.214 x 0.677 + 0.530 x 0.994 + 0.256 x 0.592 = 0.823 (0.824). The dip at 0.75 is retrace's share falling faster than the clock's share rises (an identity given the class accuracies; the explanatory content is that one layer implements these two rules and only these). Registered as "no mechanism sentence"; this is a post hoc decomposition and should be presented as such or not at all.

4.6 Dyck-2: blocks to skip (`dyck_distance.py`, `dyck_crosstab.py` -> `out/dyck_distance_out.txt`, `out/dyck_crosstab_out.txt`)

Closing accuracy (A2f's per-prefix renormalised legal-closer probability, closer-feasible prefixes of the registered L32 D12 cell) by the number of complete balanced blocks between the top opener and the prefix; mean over 8 seeds.

blocks to skip (n prefixes)	RoPE 1L	2L	3L	4L	MapWM 1L
0 (3682)	0.998	0.999	0.999	0.999	0.999
1 (5839)	0.640	0.925	0.984	0.991	0.997
2 (572)	0.641	0.778	0.885	0.934	0.993
3-4 (161)	0.613	0.760	0.860	0.925	0.993

Cross-tab (`dyck_crosstab_out.txt`): at a fixed distance of 16-31 tokens, 3 layers score 0.966 / 0.896 / 0.853 at 1 / 2 / 3-4 blocks; at a fixed 1 block, 2 layers score 0.970 / 0.948 / 0.835 at distance 2-7 / 8-15 / 16-31. Both blocks and distance matter for the index models; the path model is flat in both (0.987-0.998). Reading: a mixture of the routes, consistent with but not specific to the account. 93% of scored prefixes need at most one block, which is why three layers already reach 0.964 in aggregate. By stack depth, one layer falls from 0.88 at depth 1 to 0.60 at depth 9-10; two layers or more are nearly flat.

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5. Proposed paper text (Section 5)

Replace the paragraph "What depth buys index attention" and the closing positioning paragraph with the following (labels for the editor: the first two sentences restate registered numbers; the rest is post hoc analysis and theory and must be marked "(analysis)" in the paper as the draft already does for floors):

> **Why the rate differs by task.** A revisit asks for an earlier observation at the same cell, that is, for a key s whose moves since s sum to zero. Path integration puts the prefix sum into the rotation, so one layer compares P_t - P_s directly at any gap. Index attention sees only t - s, and must compute the predicate in its residual stream. It can do so in two ways. A global count (uniform attention over the moves, rescaled, then matched) needs only two layers in principle (Liu et al.

> 2023; Yao et al. 2021), but its precision must grow with the walk length and the target is sensitive to every move since the last visit (Hahn and Rofin 2024). A local rule needs no count but only reaches returns that span a few direction runs: undoing the last run (retrace) or, when the walk never reverses, a fixed one-lap offset (a clock). Each extra layer reaches returns across more runs. The stored models show both routes (analysis; App. F). One index layer learns exactly retrace and the clock: on the ring it scores 0.68-0.80 on retraces, 0.99-1.00 on one-lap returns once they are common, and the floor on everything else, so at p_plus 1.0, where every revisit is a one-lap return, it matches the path layer. Three index layers on the ring are flat in the number of runs a revisit spans and carry a global count in the residual stream (linear-probe R^2 0.69 for P_t / t, against 0.18 at one layer). On the torus at T = 1024 no index model carries a count (R^2 <= 0.21); four layers or the loop solve retraces (0.996 / 1.000) and decay with the number of runs spanned (0.93 / 0.98 at 3-4 runs, 0.74 / 0.65 at 5-8, 0.56 / 0.50 at 9-16), at a fixed gap as well as overall, and their last layer attends to the remembered cell for retraces (0.55-0.58 of its mass) but not at long gaps (0.00-0.01; path layer 0.81). Nearly all long-gap revisits span 17 or more runs (median 37). Depth therefore buys index attention a local reach measured in direction runs; the torus exchange rate is "more than four" because its long-range revisits lie beyond that reach and the global count is not found.

Figure suggestion (Fig. 2a companion or App. F): accuracy against runs spanned (2, 3-4, 5-8, 9-16, 17+) for index 1L, 4L, loop and path on the torus, with the ring's 1L / 2L / 3L beside it; caption numbers from Sections 4.1 and 4.5.

Two sentences for Limitations: "The explanation of the exchange rate is post hoc: the class splits, attention masses and probes were computed on stored checkpoints after the registered results were read, and the reach of route B was measured only at one and four layers on the torus. The routes are distinguished by probes and by accuracy against runs spanned, not by an intervention."

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6. Proposed GPU experiments (each with a registered prediction)

Costs from stored logs: torus T = 1024 at 900 epochs ~3.1 s/epoch per index layer (L1 50 min, L4 3.2 h per run, two jobs per GPU); ring T = 128 at 300 epochs 9 min (1 layer) to 25 min (3 layers).

E1. Run length at fixed T (decisive between routes; cheapest). Torus T = 1024, walks with run length fixed at 2, 5 or 10 moves (or drawn 1-3 / 8-10), index RoPE 4 layers and path 1 layer, trained and tested on each. Prediction (route B): the accuracy-against-runs-spanned curve of the index model is the same in every condition, so short runs (more segments per gap) lower overall accuracy and long runs raise it; route A predicts dependence on T and gap only. Path flat at 1.000. 3 conditions x 2 arms x 4 seeds: ~12 x 3.2 h + 12 x 0.8 h ~ 48 GPU-h; at T = 512 ~ 24 GPU-h. A CPU eval-only version on the stored models is Section 4.4 (distribution-shifted, so it can only suggest).

E2. Reach against depth. Torus T = 1024, index RoPE at 1, 2, 3, 4, 6, 8 layers, n = 4, all retrained in one batch (rule 12). Readout: the largest runs-spanned bin with accuracy >= 0.9, per depth. Linear reach predicts ~L + 1 runs at L layers; doubling predicts

~2^L. Decides whether any practical depth reaches the torus (long gaps need 17+). Cost: 24 layer-units per seed x 47 min x 4 seeds ~ 75 GPU-h; trimmed to {2, 4, 8} ~ 44 GPU-h.

E3. Give index attention a normaliser. Add a BOS token and Lemma 6's absolute channel log(2T - t) (or t / T) to index RoPE at 2 and 4 layers on the torus. If route A is blocked by the missing normaliser, two layers should then solve long-gap and wrap-only revisits, and probes should find P_t / t; if precision or sensitivity block it, nothing changes. 2 arms x 2 depths x 4 seeds ~ 16 runs ~ 35 GPU-h. A cheap first step: the ring at T = 128 with and without BOS at 2 layers (does the count appear at 2 instead of 3? ~3 GPU-h).

E4. The ring as a function of T. Ring p_plus 0.5, T in {128, 256, 512, 1024} at matched length, with the ring size scaled so the wrap-only share is fixed (N ~ sd of P_T), index 1-4 (6) layers, n = 4. Route A predicts the rate grows with T (precision 1/T, sensitivity ~ gap) and the count probe weakens; route B predicts the rate grows with runs spanned. With E1 this separates the two. ~60 GPU-h full; T in {128, 512} x L in {2, 3, 4, 6} ~ 15 GPU-h.

E5. Dyck at matched depth across D. Dyck at L32 / L64 with D in {4, 8, 12, 16}, index 1-4 layers: does the rate rise with D (blocks to skip) at fixed length? ~20 GPU-h. Lower priority: Dyck mixes the routes (Section 4.6).

Priority: E1 (cheap, decisive, ties to the paper's main exchange-rate claim), then E3's ring step (3 GPU-h), then E2. Each needs a pre-registration and a results-blind code check (rule 29) before launch; pilots on seeds outside the batch.

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7. Caveats

- Everything in Sections 4.1-4.6 is post hoc and eval-only; it describes stored models, all of whose index runs are STALLED at their budget. "Reach" is the reach of these budgets.
- n = 4 seeds on the torus (complete separations only in the registered contrasts); seed-to-seed spread in the 4-layer model is large on non-retrace classes (0.51-0.77).
- Segment and gap are correlated in natural walks; the fixed-gap tables and Section 4.4 reduce but do not remove that.
- Linear probes can miss a nonlinear code; the attention readout (no mass on the cell at long gaps) does not rely on them.
- The ring eval stream overlaps training batch 0 of seed-0 runs (`CANCEL_RESULTS.md` correction); only seeds 2-7 are read.
- Liu et al.'s and Yao et al.'s constructions use absolute positions or a padding token, dense state supervision (Liu), and hard or near-hard attention; our task supervises the state only through retrieval. They bound depth; they do not predict what is learned. Yao et al. (2021) is cited through Liu et al.'s App. A.5 (not in `papers/`).
- Hahn and Rofin's bounds are for Boolean functions under uniform inputs; we use them for direction only.